Global–Local Information Soft-Alignment for Cross-Modal Remote-Sensing Image–Text Retrieval

Gang Hu, Zaidao Wen, *Member, IEEE*, Yafei Lv, Jianting Zhang, and Qian Wu

***Abstract*— Cross-modal remote-sensing image–text retrieval (CMRSITR) is a challenging task that aims to retrieve target remote-sensing (RS) images based on textual descriptions. However, the modal gap between texts and RS images poses a significant challenge. RS images comprise multiple targets and complex backgrounds, necessitating the mining of both global and local information (GaLR) for effective CMRSITR. Existing approaches primarily focus on local image features while disregarding the local features of the text and their correspondence. These methods typically fuse global and local image features and align them with global text features. However, they struggle to eliminate the influence of cluttered backgrounds and may overlook crucial targets. To address these limitations, we propose a novel framework for CMRSITR based on a transformer architecture, which leverages global–local information soft alignment (GLISA) to enhance retrieval performance. Our framework incorporates a global image extraction module, which captures the global semantic features of image–text pairs and effectively represents the relationships among multiple targets in RS images. In addition, we introduce an adaptive local information extraction (ALIE) module that adaptively mines discriminative local clues from both RS images and texts, aligning the corresponding fine-grained information. To mitigate semantic ambiguities during the alignment of local features, we design a local information soft-alignment (LISA) module. In comparative evaluations using two public CMRSITR datasets, our proposed method achieves state-of-the-art results, surpassing not only traditional cross-modal retrieval methods by a substantial margin but also other contrastive language-image pretraining (CLIP)-based methods.**

***Index Terms*— Adaptive local information extraction (ALIE), cross-modal remote-sensing image–text retrieval (CMRSITR), fine-grained, global–local information soft-alignment (GLISA).**

I. Introduction

With the rapid development of Earth observation technologies, there has been a continuous increase in the number of remote-sensing (RS) images [1]. These RS images offer a vast amount of geographical, environmental, and natural resource information, which holds significant value for research and applications in various fields such as military, agriculture, forestry, and marine sciences [2], [3]. However, before one can explore and use this information, the challenge lies in efficiently obtaining the required RS images from the enormous volume of RS big data. To tackle this challenge, researchers commonly use retrieval methods that enable the automatic acquisition of the desired RS images [4], [5].

RS image retrieval methods can be broadly categorized into single-modal and cross-modal RS image retrieval. Single-modal image retrieval methods focus on searching images with similar content to a query image from a set of candidate images of the same modality [6], [7], [8], [9], [10]. Alternatively, cross-modal RS image retrieval involves retrieval across different modalities, including image–text retrieval [11], [12], [13], image–audio retrieval [14], [15], [16], and optical–SAR image retrieval [17], [18]. Among the various modalities, cross-modal RS image–text retrieval (CMRSITR) has gained increasing attention. This task aims to retrieve relevant RS images based on textual queries, as textual descriptions can provide valuable and detailed information about the targets and their relationships in the RS scene. In addition, textual data are often more readily available and accessible compared with RS images. In recent years, deep learning techniques have been widely adopted in CMRSITR methods to extract features from both RS images and texts. These features are then mapped into a joint embedding space, where similarity or matching scores are computed. Given the modality heterogeneity between images and texts, the key to improving the retrieval performance in CMRSITR is to extract discriminative feature representations. Based on the type of image feature extractor used, CMRSITR methods can be mainly categorized into deep convolutional neural network (DCNN) [19]-based methods and transformer [20]-based methods. DCNN-based methods, such as asymmetric multimodal feature matching network (AMFMN) [13] and global and local information (GaLR) [12], typically use DCNNs to extract RS image feature representations and recurrent neural networks (RNNs) [21] to extract text feature representations. These methods often optimize the model using a bidirectional triplet loss [22]. Transformer-based methods, such as interacting-enhancing feature transformer (IEFT) [11], leverage transformer architectures to extract feature representations for both images and texts.

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A. Motivation

However, a common drawback of the existing CMRSITR methods is their lack of explicit focus on extracting and aligning local features in both image and text modalities. Specifically, GaLR [12] and IEFT [11] primarily address the local features of the RS image but still align the global features of image–text pairs. Although some cross-modal natural scene image–text retrieval methods [23], [24] consider the local information, they often require the pretrained Faster region-based convolutional neural network (R-CNN) [25] to extract region features in advance. Given that RS images and text descriptions often encompass multiple targets and complex backgrounds, solely aligning global features proves inadequate in effectively eliminating redundant background interference and may even disregard crucial targets, resulting in unsatisfactory retrieval outcomes. Hence, the alignment of local features in image–text pairs becomes crucial for achieving accurate retrieval. By prioritizing the extraction and alignment of local features, we can better capture fine-grained correspondences between RS images and texts, eliminate redundant background interference, and effectively emphasize key target features, thereby yielding more precise retrieval results. However, a major challenge lies in the current labeling of CMRSTIR datasets, which primarily exists at the instance level (image level, sentence level) rather than providing the fine-grained (region level, word level) information required to mine and align such details.

With the rapid advancement of large-scale vision–language pretraining (VLP) models, such as CLIP [26] and vision-and-language transformer (ViLT) [27], substantial progress and performance enhancements have been achieved in cross-modal image–text tasks. Among these models, the most notable one is CLIP, which uses contrastive language–image pretraining. VLP models effectively integrate visual and language information and undergo pretraining on extensive datasets to capture semantic alignment between images and texts, enabling the acquisition of comprehensive visual and language representations. However, the application of VLP models in the CMRSITR task is still in its nascent stages. DCNN-based CMRSITR methods initialize their model parameters using separate pretrained models on unimodal datasets, while transformer-based methods use VLP models for initialization, equipping the models with the underlying alignment capability required to handle multimodal data. Nonetheless, as illustrated in Fig. 1, the targets in natural images are very significant and few in number, while more targets and more complex backgrounds exist in RS images. Therefore, it is impractical to directly use these VLP models in the CMRSITR task to achieve better retrieval performance due to the task gap. Transferring the powerful image–text knowledge of VLP models to the CMRSITR domain and designing high-performance cross-modal retrieval methods specifically addressing the challenges posed by a large number of targets and complex backgrounds in this domain is a challenging task.

B. Contributions

To address the aforementioned challenges, we propose a novel framework called global–local information soft-alignment (GLISA) for the CMRSITR task. Unlike previous approaches that primarily focus on local image information, our framework incorporates both instance-level alignment and fine-grained alignment (refer to Fig. 1). The GLISA model comprises three key modules: the global information extraction (GIE) module, the adaptive local information extraction (ALIE) module, and the local information soft-alignment (LISA) module. The GIE module includes an image encoder and a text encoder, which extract global semantic features from images and texts, respectively. These features are then aligned using an image–text contrastive (ITC) loss, facilitating cross-modal alignment at the global level. The ALIE module dynamically extracts local features that correspond to both the image and the text. This adaptive extraction allows the model to focus on more fine-grained and context-aware information, enhancing the alignment process. To address semantic ambiguities that may arise during the local alignment process, we introduce the LISA module. This module aims to facilitate soft alignment by considering the relationships between local image regions and words in the text. By leveraging these relationships, the LISA module helps mitigate potential ambiguities and improve the overall alignment quality. We conduct extensive experiments on the RS image captioning data (RSICD) and RS image–text match dataset (RSITMD) datasets for CMRSITR. The experimental results demonstrate that GLISA outperforms state-of-the-art methods by a wide margin. The main contributions of our work can be summarized as follows.

1) We propose an end-to-end GLISA model that addresses the challenges of cross-modal alignment in CMRSITR. Unlike previous approaches that primarily focus on global alignment, GLISA goes a step further by aligning

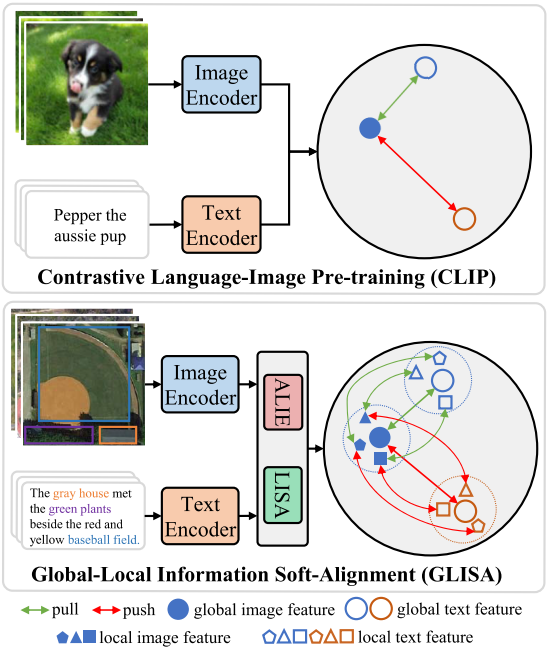

Fig. 1. Motivation for the proposed GLISA framework. Unlike the natural image that contains significant target, the RS image contains multiple small targets and cluttered background, so the VLP (CLIP) model cannot be directly used for CMRSITR. Our GLISA method performs not only global image–text alignment but also local image–text alignment.

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both GaLR. During the training process, GLISA aligns global information between image–text pairs while also mining local information from regions and words to align them. This comprehensive alignment strategy enables GLISA to capture both the overall context and the fine-grained details, allowing it to pay more attention to fine-grained information during the inference process. By considering both global and local alignment, GLISA offers a more comprehensive and effective solution for CMRSITR.

2) To address the challenge of semantic ambiguity in cross-modal alignment, we introduce the LISA module. Even in cases where image–text pairs are mismatched, there may still be matching local regions and words. The LISA module is designed to alleviate semantic ambiguity by considering the relationships between these local regions and words. By leveraging these relationships, LISA further improves the retrieval performance by mitigating the impact of semantic ambiguities. This novel module enhances the robustness of the GLISA model and contributes to its superior performance in CMRSITR tasks.
3) In contrast to traditional approaches that use a bidirectional triplet loss, GLISA uses the ITC loss to narrow the modal gap. The ITC loss does not require presetting a margin parameter, making it more flexible and adaptable to different retrieval scenarios. By leveraging the ITC loss, GLISA enhances the alignment between image and text modalities, effectively bridging the gap between them.

The rest of this article is organized as follows. In Section II, we briefly review previous work related to CMRSITR and the VLP model. Then, we provide a detailed description of the proposed GLISA model and its components in Section III. In Section IV, quantitative and qualitative experimental results demonstrate the effectiveness of the proposed model. Finally, we summarize our work in Section V.

II. Related Work

In this section, we begin by reviewing the previous work related to CMRSITR to further emphasize the motivation behind this research. Subsequently, we introduce several VLP models that have inspired the development of our proposed framework.

A. Cross-Modal Remote-Sensing Image–Text Retrieval

With the rapid development of RS technologies, the availability of RS images has significantly increased. Consequently, CMRSITR has emerged as a prominent research topic due to its potential for retrieving relevant images using readily accessible text. The key challenge lies in effectively mapping and aligning image–text features from different modalities into a unified embedding space. To address this challenge, several approaches have been proposed. Abdullah et al. [28] introduced a deep bidirectional triplet network (DBTN) that leverages an average fusion strategy to combine features associated with five image sentences. This approach enhances feature robustness and facilitates more effective learning. Yuan et al. [13] tackled the problem of multiscale scarcity and target redundancy by proposing an AMFMN. By extracting multiscale information from RS images and dynamically filtering redundant features, AMFMN improved the retrieval performance. In addition, Yuan et al. released a fine-grained RSITMD to facilitate research in this domain. To reduce computational costs during inference, Yuan et al. [29] developed a concise yet effective cross-modal retrieval model called lightweight RS multiscale cross-modal retrieval model (LW-MCR). This model focuses on multiscale information and dynamically filters redundant features. In addition, they introduced a knowledge distillation-based hidden supervised optimization method to further enhance retrieval performance. Recognizing the importance of local features that capture target relationships and saliency, Yuan et al. [12] proposed a CMRSITR framework based on global–local information. They designed a multilevel information dynamic fusion (MIDF) module that integrates global and local features, resulting in more salient representations. Furthermore, they introduced a plug-and-play multivariate rerank (MR) algorithm to further improve retrieval performance. To reduce the reliance on labeled training samples, Mikriukov et al. [30] proposed a deep unsupervised cross-modal contrastive hashing (DUCH) method. By leveraging unsupervised learning, DUCH enables effective retrieval without the need for extensive labeled data.

The aforementioned methods commonly use DCNNs to extract image features and RNNs to extract text features. Model parameters are typically initialized using separate pretrained models on unimodal datasets. Due to the self-attention mechanism within the transformer architecture, transformer has gained widespread attention. More and more transformer-based methods have been proposed, which use transformer architectures to extract features of images and texts. Rahhal et al. [31] introduced a contrasting dual-transformer architecture for the CMRSITR task. To enhance the communication with the retrieval system, Rahhal et al. [32] proposed a multilanguage framework based on transformers. To explore potential correlations between RS images and texts, Li et al. [33] introduced a fusion-based contrastive learning model (FBCLM) for CMRSITR that uses the self-attention module and the cross-modal fusion module to extract discriminative features. FBCLM can retrieve RS images in both Chinese and English. Wang et al. [34] proposed a multiscale interactive transformer (MSIT) model to discover the potential relationship between different modalities and learn the multiscale features of RS images. Tang et al. [11] proposed an IEFT model for CMRSITR, which treats RS images and text as a unified input to the model. IEFT uses the retrieval module (RM) to output the matching scores and sort them to get the final retrieval results. In addition to FBCLM, other transformer-based methods all use the VLP (i.e., CLIP [26], ViLT [27]) models to initialize the model parameters, enhancing the cross-modal underlying alignment of the model. These methods also demonstrate that the VLP models can be applied to CMRSITR and has better retrieval performance. To reduce the training parameters and improve

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the training efficiency, Yuan et al. [35] proposed a parameter-efficient transfer learning (PETL) framework, which contains a pretrained CLIP model, a multimodal RS adapter, and a hybrid multimodal contrastive (HMMC) learning objective.

Reviewing these CMRSITR methods, although GaLR and IEFT focus on mining local information in RS images and demonstrate that this information can effectively improve retrieval performance, they have neglected the local information of texts and the correspondence between local information of RS images and texts. Since RS images and texts contain multiple targets and redundant information, focusing on global information alignment or only local information of the RS image may ignore some important targets and affect the retrieval results. In this work, our GLISA method makes full use of GaLR of image–text pairs to mine discriminative features and fine-grained clues, which effectively improves the retrieval performance.

B. Vision–Language Pretraining Model

VLP models have demonstrated their ability to capture rich semantic relationships and enhance the understanding of visual scenes by leveraging contextual textual information through pretraining on large-scale image–text pairs. These models have shown remarkable performance on various downstream tasks, including image–text retrieval [36], image captioning [37], visual entailment [38], visual question answering [39], natural language visual reasoning [40], and visual grounding [41]. VLP models can be broadly categorized into single-stream models and dual-stream models. Single-stream models, such as visual bidirectional encoder representation from transformers (VisualBERT) [42], Uniter [43], Oscar [44], and ViLT [27], take the visual and text embeddings as a whole and pass them through a self-attention transformer encoder. These models output similarity scores for the image–text pairs, capturing potential relationships between the two modalities. However, these single-stream models can be computationally expensive during image–text retrieval due to the large input size and complex attention mechanisms. On the other hand, dual-stream models, including CLIP [26], a large-scale image and noisy-text embedding (ALIGN) [45], and multimodal end-to-end transformer (METER) [46], use separate encoders for images and texts. These models independently encode the image and text features and then map them into a joint embedding space. Dual-stream models consume less time during image–text retrieval while achieving significant performance. CLIP, in particular, has gained popularity among dual-stream models. It uses the information noise-contrastive estimation (InfoNCE) loss [47] to optimize the model and align the features of image–text pairs. Due to the fact that the image–text pairs used by CLIP are obtained through web crawling and are not of high quality, significant noise exists. As a result, the image pairs are not simple one-to-one relationships, but complex many-to-many relationships [48]. Therefore, several methods, such as PyramidCLIP [49] and SoftCLIP [50], have been proposed to mitigate this issue.

Due to the task gap between natural image–text retrieval and CMRSITR, applying CLIP directly to the RS domain for CMRSIT is suboptimal. The characteristics and challenges specific to RS data, such as multiple targets, nonprominent targets, complex and redundant backgrounds, and domain-specific semantics, require tailored approaches for effective CMRSITR. Therefore, directly applying CLIP, which is not specifically designed for the RS domain, may not yield optimal results in the CMRSITR task. Our GLISA method effectively transfers the rich image–text knowledge of CLIP to the CMRSITR task. Considering the characteristics of multiple targets and redundant information in CMRSITR, we explore fine-grained clues corresponding to image–text pairs to enhance retrieval performance.

III. Proposed Framework

In this section, we present the GLISA framework proposed for CMRSITR. Fig. 2 illustrates the overview framework of GLISA that includes a GIE module, an ALIE module, and an LISA module. More details are discussed in Sections III-A–III-E.

A. Global Information Extraction

As depicted in Fig. 2, the GIE module comprises an image encoder and a text encoder, both designed as transformer architectures. These encoders are specifically engineered to extract global information from the respective modalities. Using transformer-based encoders for both the image and text modalities, the GIE module adeptly extracts global information from both the sources. This comprehensive global information serves as a foundational element for subsequent processing and alignment steps within the cross-modal retrieval framework.

1) Image Encoder: For an input image $V \in \mathbb{R}^{H\times W\times C}$, we can obtain its embeddings using the patch embedding block, as shown in Fig. 3. Specifically, the image is first divided into a sequence of flattened 2-D patches $V_p \in \mathbb{R}^{N\times(P^2\cdot C)}$, where (H, W) denotes the size of the input image, C is the number of channels, P is the patch size, $N = HW/P^2$ is the number of patches, and then flatten and map the patch sequence to 1-D tokens $E^v = \{e_1^v, \ldots, e_N^v\} \in \mathbb{R}^{N\times D^v}$ by a trainable linear projection. Next, a [CLS] token $e_{\text{cls}} \in \mathbb{R}^{D^v}$ is appended to the token sequence and a learnable positional embedding $E_{\text{pos}}^v \in \mathbb{R}^{(N+1)\times D^v}$ is added. The resulting embedded sequence of patches that serves as input to the image encoder is

$$F_0^v = \left\{e_{\text{cls}}^v, e_1^v, \ldots, e_N^v\right\} + E_{\text{pos}}^v. \tag{1}$$

The image encoder is composed of an L-layer transformer architecture, where each transformer layer consists of a multihead self-attention (MSA) block and a multilayer perceptron (MLP) block. The MSA block concatenates the features outputted by each head and passes this concatenated feature through a linear layer. The MLP block contains two fully connected layers with a Gaussian error linear unit (GELU) nonlinearity. In both the blocks, Layernorm (LN) is applied first, followed by the addition of the input and the output of the block using residual connections. The overall process is formulated as

$$\begin{aligned} F_{l,\text{msa}}^v &= \text{MSA}\left(\text{LN}\left(F_{l-1}^v\right)\right) + F_{l-1}^v, \quad l = 1, \ldots, L \\ F_l^v &= \text{MLP}\left(\text{LN}\left(F_{l,\text{msa}}^v\right)\right) + F_{l,\text{msa}}^v, \quad l = 1, \ldots, L. \end{aligned} \tag{2}$$

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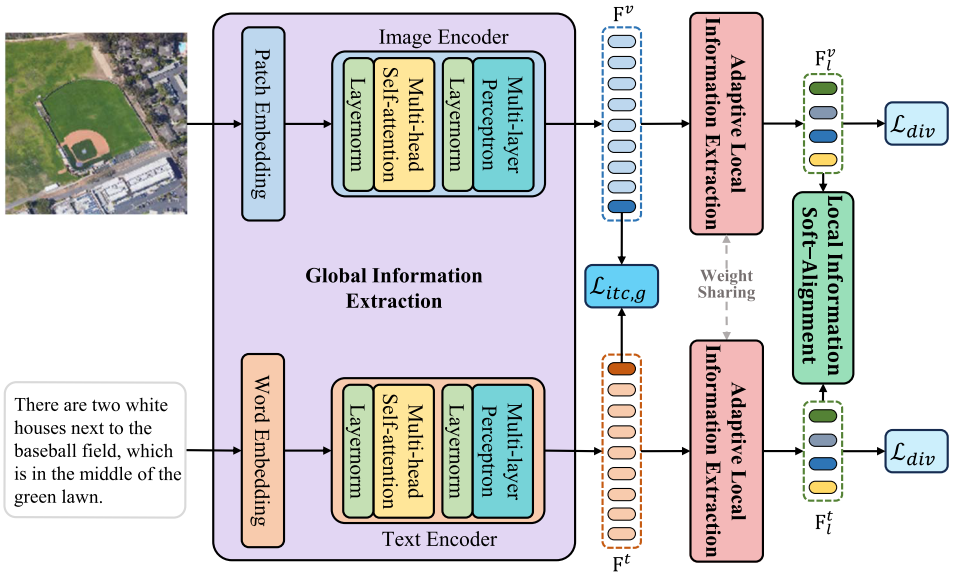

Fig. 2. Overview of the proposed GLISA framework. It is a dual-stream model, which contains a GIE module, an ALIE module, and an LISA module. The purpose of GIE is to extract global features and to align global semantic information. ALIE aims to adaptively extract local features of image–text pairs by the MSA. The goal of LISA is to reduce ambiguities that may arise during the process of aligning local information.

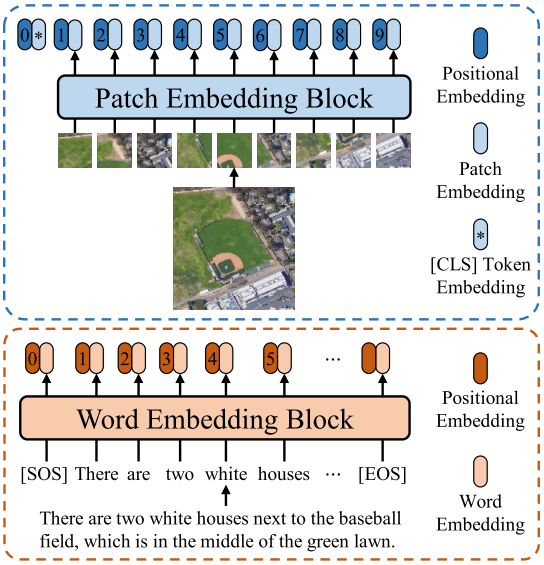

Fig. 3. Illustration of the patch embedding block and the word embedding block. Before the image encoder, the patch embedding block converts local regions of RS image into vector representations. The purpose of the word embedding block is to convert the text into vector representations.

Finally, we adopt a linear projection to map the output F_L^v of the image encoder to the image–text joint embedding space, resulting in the final image feature sequence $F^v = \{f_{\text{cls}}^v, f_1^v, \ldots, f_N^v\} \in \mathbb{R}^{(N+1)\times D}$, where D is the embedding dimension, f_{cls}^v is the global feature of the input image and is denoted as f_g^v, and $\{f_1^v, \ldots, f_N^v\}$ are patch features.

2) Text Encoder: As illustrated in Fig. 3, the embeddings for an input text T are obtained using the word embedding block. The process involves several steps. First, we apply lower cased byte pair encoding (BPE) with a vocabulary size of 49 152 to convert each word into a unique numeric ID [51]. Subsequently, these IDs are mapped to D^t-dimensional word embeddings, resulting in a text sequence denoted as $E^t = \{e_1^t, \ldots, e_M^t\} \in \mathbb{R}^{M\times D^t}$. To indicate the start and end of the text, special tokens [SOS] (start of sentence) and [EOS] (end of sentence) are appended to the text sequence. In addition, a learnable positional embedding $E_{\text{pos}}^t \in \mathbb{R}^{(M+2)\times D^t}$ is added to represent the order of the text. The resulting text embedding, which serves as the input for the text encoder, is obtained through these aforementioned steps. The final text embedding used as input for the text encoder is

$$F_0^t = \{e_{\text{sos}}^t, e_1^t, \ldots, e_M^t, e_{\text{eos}}^t\} + E_{\text{pos}}^t. \tag{3}$$

In accordance with CLIP, the text encoder incorporates a modified transformer architecture described by Radford et al. [26], which is similar to the image encoder. The text encoder uses masked multiheaded self-attention to capture the correlations among each word within the text. The initial text embedding, denoted as F_0^t, is passed through the text encoder, resulting in the output F_L^t. Subsequently, the output of the text encoder is linearly projected in the image–text joint embedding space to derive the final text feature sequence $F^t = \{f_{\text{sos}}^t, f_1^t, \ldots, f_M^t, f_{\text{eos}}^t\} \in \mathbb{R}^{(M+2)\times D}$. Within this sequence, f_{eos}^t represents the global feature of the input text and is denoted as f_g^t. The features $\{f_1^t, \ldots, f_M^t\}$ correspond to individual word features.

Given a mini-batch of B image–text pairs $\{(V,T)\}_{i=1}^B$, we input them into the GIE module to obtain image–text global feature pairs $\{(f_g^v, f_g^t)\}_{i=1}^B$. For the ith pair, the image-to-text matching probability vector $p_i(V,T) = \{p_{i,j}(V,T)\}_{j=1}^B$ is calculated through

$$p_{i,j}(V,T) = \frac{\exp\left(\text{sim}\left(f_{g,i}^v, f_{g,i}^t\right)/\tau\right)}{\sum_{j=1}^B \exp\left(\text{sim}\left(f_{g,i}^v, f_{g,j}^t\right)/\tau\right)} \tag{4}$$

and the text-to-image matching probability vector $p_i(T,V) = \{p_{i,j}(T,V)\}_{j=1}^B$ is

$$p_{i,j}(T,V) = \frac{\exp\left(\text{sim}\left(f_{g,i}^t, f_{g,i}^v\right)/\tau\right)}{\sum_{j=1}^B \exp\left(\text{sim}\left(f_{g,i}^t, f_{g,j}^v\right)/\tau\right)} \tag{5}$$

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where $\mathrm{sim}(\cdot,\cdot)$ computes the cosine similarity, and τ is a learnable temperature parameter controlling the smoothness or sharpness of the probability distribution. Then the ITC loss and the text-to-image contrastive loss functions are, respectively, computed as

$$\mathcal{L}_{\mathrm{i2t},g} = \frac{1}{B}\sum_{i=1}^{B} H(q_i(V,T),\, p_i(V,T)) \tag{6}$$

$$\mathcal{L}_{\mathrm{t2i},g} = \frac{1}{B}\sum_{i=1}^{B} H(q_i(T,V),\, p_i(T,V)) \tag{7}$$

where $H(\cdot,\cdot)$ denotes the cross-entropy function, $q_i(V,T) = \{q_{i,j}(V,T)\}_{j=1}^{B}$ and $q_i(T,V) = \{q_{i,j}(T,V)\}_{j=1}^{B}$ are the real matching probability vector for the ith image–text pair, $q_{i,j} = y_{i,j}/\sum_{k=1}^{B} y_{i,k}$, $y_{i,j} = 1$ means the pair (V_i, T_j) is matched, while $y_{i,j} = 0$ indicates the unmatched pair. Then we can train the GIE module using the ITC loss $\mathcal{L}_{\mathrm{itc},g}$ to align global features. The ITC loss not only helps the model learn the relationships between positive sample pairs but also assists in learning the relationships between negative sample pairs

$$\mathcal{L}_{\mathrm{itc},g} = (\mathcal{L}_{\mathrm{i2t},g} + \mathcal{L}_{\mathrm{t2i},g})/2. \tag{8}$$

B. Adaptive Local Information Extraction

In the case of RS images, although the global feature can capture the overall semantic information, these images often contain important target-specific details. To address this, we extract local features that specifically represent the small-scale target information within the RS image. In contrast to previous approaches that solely extracted local features from the image, we extract local features from both the image and the accompanying text, and align them. Patch features and word features are used to represent the local features of the image and text, respectively. However, these features may contain irrelevant background noise, and the alignment relationships between patches and words are not known beforehand. To address these challenges, we use MSA [52]. Due to the ability of MSA to allow the model to focus on different parts of the input and learn diverse representations, we use the features outputted by each head to represent local information. Using MSA, we can adaptively extract local features from both the image and text, taking into account their inherent alignments.

1) Image Local Information Extraction: We take the image feature sequence F^v as input to the ALIE module and output K image local features $F_l^v = \{f_{l,1}^v, \ldots, f_{l,K}^v\} \in \mathbb{R}^{K\times D}$, where K is the number of heads. As shown in Fig. 4, for the ith head ($i = 1, \ldots, K$), we first input the sequence F^v into LN and linearly project it into three vectors: query $\mathcal{Q}_i$, key $\mathcal{K}_i$, and value $\mathcal{V}_i$

$$\mathcal{Q}_i = F^v W_i^{\mathcal{Q}}, \quad \mathcal{K}_i = F^v W_i^{\mathcal{K}}, \quad \mathcal{V}_i = F^v W_i^{\mathcal{V}} \tag{9}$$

where $W_i^{\mathcal{Q}}, W_i^{\mathcal{K}}, W_i^{\mathcal{V}} \in \mathbb{R}^{D\times D}$ are the trainable parameters. Then we obtain the output of the ith head as

$$\widehat{F}_i^v = \mathrm{softmax}\left(\frac{\mathcal{Q}_i \mathcal{K}_i^{T}}{\sqrt{D}}\right)\mathcal{V}_i \tag{10}$$

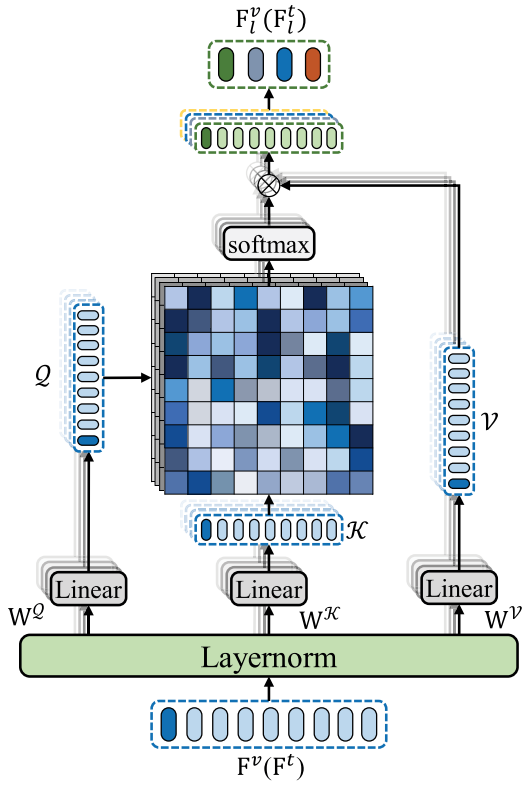

Fig. 4. Illustration of the ALIE module. The ALIE module uses the MSA to extract diverse local features. The output of each head represents the local information of the image (or text).

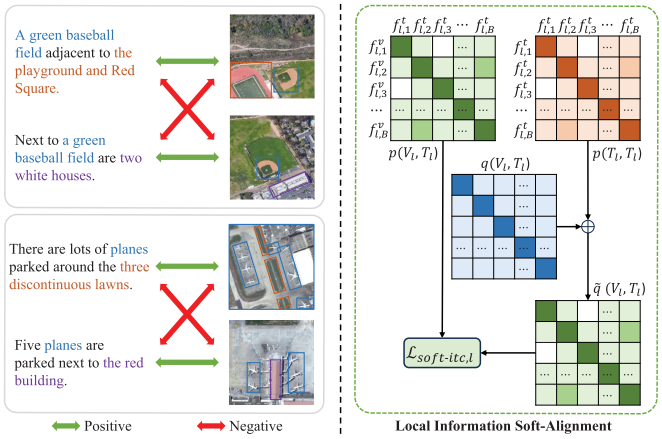

Fig. 5. (Left) Different image–text pairs have similar local regions and the same words, and thus are prone to semantic ambiguity in the process of local information alignment by the traditional contrastive loss $\mathcal{L}_{\mathrm{itc},l}$. (Right) Illustration of the LISA module. It softens the hard true matching probability vectors and reduces semantic ambiguity in the training process.

where $\widehat{f}_i^v = \widehat{F}_i^v(0,:) \in \mathbb{R}^D$ represents the global feature output by the ith head. Since different heads focus on different parts, the global feature $\widehat{f}_i^v$ of the ith head can represent the ith local feature of the image, denoted as $f_{l,i}^v$. Therefore, we can obtain K local features $\{f_{l,1}^v, \ldots, f_{l,K}^v\}$ of the input image by different heads.

2) Text Local Information Extraction: Since the text feature sequence F^t and the image feature sequence F^t are already in the image–text joint embedding space, we use the

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parameter-shared ALIE module to extract text local features $F_l^t = \{f_{l,1}^t, \ldots, f_{l,K}^t\} \in \mathbb{R}^{K\times D}$.

To optimize the parameters of the ALIE module, we also use the ITC loss $\mathcal{L}_{\text{itc},l}$ to achieve cross-modal alignment of local features

$$\begin{aligned}\mathcal{L}_{\text{itc},l} &= (\mathcal{L}_{\text{i2t},l} + \mathcal{L}_{\text{t2i},l})/2 \\ &= \left(\sum_{j=1}^{K}\mathcal{L}_{\text{i2t},l,j} + \sum_{j=1}^{K}\mathcal{L}_{\text{t2i},l,j}\right)\Big/2. \end{aligned} \tag{11}$$

To ensure that the local features extracted from both the image and text are diverse and focus on different parts, we introduce a diversity loss term denoted as $\mathcal{L}_{\text{div}}$. The purpose of this loss term is to constrain the ALIE module and encourage the local features to be distinct from each other. By incorporating the diversity loss, we aim to reduce redundancy among the local features. This means that each local feature will capture unique and nonrepetitive information, enhancing the overall representation of the image and text. The diversity loss term encourages the ALIE module to learn diverse attention patterns, enabling the local features to focus on different parts or aspects of the input

$$\mathcal{L}_{\text{div}} = \frac{1}{K(K-1)}\sum_{i=1}^{K}\sum_{j=1,j\neq i}^{K}\left(\text{sim}\left(f_{l,i}^v, f_{l,j}^v\right)+\text{sim}\left(f_{l,i}^t, f_{l,j}^t\right)\right). \tag{12}$$

C. Local Information Soft-Alignment

In Fig. 5 (left), it can be observed that different image–text pairs may share similar local regions and contain the same words. As a result, some of the local features extracted by the ALIE module can exhibit similarities as well. However, when using contrastive loss to optimize the alignment of these local features, there is a potential problem of semantic ambiguity. This arises because contrastive loss treats the matching of images and texts as a binary classification task. To address this issue, one straightforward approach is to use label smoothing [53], which involves softening the hard one-hot targets to introduce a small amount of uncertainty into the training objective. For the ith image–text pair, its calculation is

$$\tilde{q}_i = (1-\alpha)q_i + \frac{\alpha}{B-1}(\mathbb{1} - q_i) \tag{13}$$

where $\mathbb{1}$ is the all-ones vector, and α is the hyperparameter that controls the degree of smoothing. Unfortunately, despite the potential benefits of label smoothing in providing confidence to images or texts that exhibit local similarity to the ith image–text pair, it is limited in its ability to distinguish between similar and dissimilar instances. It is widely recognized that local features in different images, which are similar to the same text local feature, tend to exhibit similarity, and vice versa. To address this limitation and further enhance the alignment process, we propose leveraging the self-similarity of local features within a single modality to modulate the confidence estimation and refine the confidence assigned to individual local features. This approach recognizes that local features within the same modality should demonstrate a certain degree of resemblance. Consequently, by considering the self-similarity, we can adjust the confidence levels more effectively. Specifically, we assign higher certainty to local features that align well within the modality, while reducing the confidence for features that deviate from this intramodality consistency.

Specifically, for the ith image–text pair (V_i, T_i), we can obtain K local feature pairs $\{(f_{l,i,k}^v, f_{l,i,k}^t)\}_{k=1}^K$ by the ALIE module. Then intramodal local self-matching probability vectors of the kth local-feature pair as $p_{i,k}(V_l, V_l) = \{p_{i,j,k}(V_l, V_l)\}_{j=1}^B$ and $p_{i,k}(T_l, T_l) = \{p_{i,j,k}(T_l, T_l)\}_{j=1}^B$ can be calculated as

$$p_{i,j,k}(V_l, V_l) = \frac{\exp\left(\tilde{\text{sim}}\left(f_{l,i,k}^v, f_{l,j,k}^v\right)/\tau\right)}{\sum_{j=1}^B \exp\left(\tilde{\text{sim}}\left(f_{l,i,k}^v, f_{l,j,k}^v\right)/\tau\right)} \tag{14}$$

$$p_{i,j,k}(T_l, T_l) = \frac{\exp\left(\tilde{\text{sim}}\left(f_{l,i,k}^t, f_{l,j,k}^t\right)/\tau\right)}{\sum_{j=1}^B \exp\left(\tilde{\text{sim}}\left(f_{l,i,k}^t, f_{l,j,k}^t\right)/\tau\right)} \tag{15}$$

where $\tilde{\text{sim}}(\cdot,\cdot)$ represents that if the cosine similarity is greater than β, it is preserved; if it is smaller, it is set to 0. To enhance the alignment between images and texts, we use the text local self-matching probability vectors to soften the image-to-text true matching probability vectors [as shown in Fig. 5 (right)]. Similarly, we use the image local self-matching probability vectors to smooth the text-to-image true matching probability vectors. The formulations are expressed as

$$\tilde{q}_{i,k}(V_l, T_l) = (1-\alpha)q_i(V_l, T_l) + \frac{\alpha}{B-1}p_{i,k}(T_l, T_l) \tag{16}$$

$$\tilde{q}_{i,k}(T_l, V_l) = (1-\alpha)q_i(T_l, V_l) + \frac{\alpha}{B-1}p_{i,k}(V_l, V_l). \tag{17}$$

Next, for the kth image–text local feature pair, the soft local ITC and text-to-image contrastive loss are calculated as

$$\mathcal{L}_{\text{soft-i2t},l,k} = \frac{1}{B}\sum_{i=1}^{B}\text{H}(\tilde{q}_{i,k}(V_l, T_l), p_{i,k}(V_l, T_l)) \tag{18}$$

$$\mathcal{L}_{\text{soft-t2i},l,k} = \frac{1}{B}\sum_{i=1}^{B}\text{H}(\tilde{q}_{i,k}(T_l, V_l), p_{i,k}(T_l, V_l)). \tag{19}$$

We can obtain the image–text local contrastive loss by

$$\begin{aligned}\mathcal{L}_{\text{soft-itc},l} &= (\mathcal{L}_{\text{soft-i2t},l} + \mathcal{L}_{\text{soft-t2i},l})/2 \\ &= \left(\sum_{k=1}^{K}\mathcal{L}_{\text{soft-i2t},l,k} + \sum_{j=1}^{K}\mathcal{L}_{\text{soft-t2i},l,k}\right)\Big/2. \end{aligned} \tag{20}$$

D. Training and Inference

During the training phase, our GLISA model not only extracts global features for semantic alignment but also captures local features for target alignment, which can further improve the retrieval performance. The GLISA model is trained in an end-to-end manner, and the overall loss is defined as

$$\begin{aligned}\mathcal{L} &= \mathcal{L}_{\text{itc}} + \lambda\mathcal{L}_{\text{div}} \\ &= \frac{1}{1+K}(\mathcal{L}_{\text{itc},g} + \mathcal{L}_{\text{soft-itc},l}) + \lambda\mathcal{L}_{\text{div}}. \end{aligned} \tag{21}$$

In the inference phase, for the text (or image) queries and image (or text) candidates, we first obtain their global features.

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Then, we compute the similarity between the image–text pair for cross-modal retrieval.

E. Discussion

To address the challenge of suboptimal retrieval performance caused by multiple targets and redundant information in RS images and text, we have developed the GLISA framework for CMRSITR in Sections III-A–III-D. The key strength of our framework lies in its ability to align both global semantic information and local fine-grained information in image–text pairs. First, the proposed GIE module is introduced to extract global semantic features and align them. By effectively leveraging the remarkable cross-modal knowledge of CLIP, we successfully transfer this knowledge to the CMRSITR domain. This integration of cross-modal knowledge not only enhances the retrieval performance but also ensures the utilization of state-of-the-art techniques. Second, we design the ALIE module to capture fine-grained information. Unlike previous methods such as GaLR [12], which require an additional pretrained object detector to extract objects in RS images, our ALIE module can adaptively extract local RS image features corresponding to the local text features. This adaptive approach eliminates the reliance on external detectors and streamlines the retrieval process, resulting in improved efficiency and accuracy. Furthermore, we observe that the local alignment process involves complex relationships between local features in the image and text. These relationships are not simple one-to-one mappings but rather many-to-many relationships, which can introduce semantic ambiguity. To tackle this challenge, we propose the LISA module, which enhances retrieval performance by effectively managing the many-to-many relationships and reducing semantic ambiguity, resulting in more accurate and meaningful retrieval results. Our framework not only incorporates the use of prevalent VLP models for CMRSITR but also provides tailored solutions specifically designed to address the unique challenges of this task.

IV. Experiments and Analysis

In this section, we first introduce the experiment settings, which includes datasets, evaluation metrics, and implementation details. Next, we compare the GLISA model with other methods and conduct ablation experiments to validate the effectiveness of our proposed method. Finally, we present qualitative results to further illustrate the importance of local feature alignment for CMRSITR.

A. Experimental Settings

1) Datasets: We will evaluate the proposed GLISA model on two widely recognized datasets for CMRSITR: RSICD [61] and RSITMD [13]. The RSICD dataset is a large-scale RS image caption dataset that has been compiled from various sources such as Google Earth, Baidu Map, MapABC, and Tianditu. It consists of 10 921 RS images representing 30 different scenes. The RS images in the dataset have fixed dimensions of 224×224 pixels, but they exhibit variations in terms of resolution. Each RS image in RSICD is associated with five corresponding sentences that provide textual descriptions. The RSITMD dataset serves as a refinement and extension of the RSICD dataset, offering more fine-grained captions. It comprises a total of 4743 RS images that are categorized into 32 different classes. Similar to RSICD, each RS image in RSITMD is accompanied by five textual descriptions. The image size in the RSITMD dataset is 256×256 pixels.

2) Evaluation Metrics: To evaluate our GLISA model, we adopt two widely used metrics $R@k$ (where $k = 1, 5, 10$) and mR as the main evaluation metrics. $R@k$ represents the probability of the ground truth appearing in the top-k candidate (image or text) list when given a query (text or image). mR is the average of six $R@k$s, which reports the overall performance of the model.

3) Implementation Details: All our experiments are conducted on a single NVIDIA RTX4090 24-GB GPU. Following the data split of Tang et al. [11], we use 90% of the dataset as the training set and 10% of the dataset as the testing set to compare with the traditional cross-modal retrieval methods. In addition, we also follow the data split of Yuan et al. [13], where 80% of the dataset is used as the training set, 10% of the dataset is used as the test set, and 10% of the dataset is used as the test set for comparing with the CLIP-based methods. For faster model convergence, the GIE module is initialized by the pretrained parameters of CLIP, and the ALIE module is randomly initialized. For the GIE module, the image encoder and the text encoder contains 12 transformer layers separately. The hidden size D^v of the image encoder is set to 768, and the hidden size D^t of the text encoder is set to 512. The dimension of the image–text joint embedding space is also set to 512. For the ALIE module, the hidden size and number of heads are set to 512 and 8, respectively.

In the training process, we use random cropping and random rotation to augment the input RS images. Before being fed into the GLISA model, the RS images are resized to 224×224. For parallel computation, the length of the text sequence is fixed to 77. We use the Adam optimizer to train the GLISA model for 20 epochs and set the batch size B to 128. The initial learning rate of the GIE module is set to $5e^{-6}$, and the initial learning rate of the ALIE module is set to $5e^{-4}$. A cosine schedule is used to decay the learning rate. The hyperparameters α and β are set to 0.1 and 0.9, respectively. The loss balance factor λ is set to 0.3.

B. Comparison With State-of-the-Art Methods

In this section, we compare the retrieval performance of our GLISA model with the traditional cross-modal retrieval methods and CLIP-based methods.

1) Traditional Cross-Modal Retrieval Methods: Following [11], we compare our method with the following traditional methods on two CMRSITR datasets.

1) *AMFMN [13]:* AMFMN use the multiscale visual self-attention (MVSA) module and the visual-guided attention (VGA) module to extract the salient features and reduce redundant features. According to different VGA modules, we compare my model with AMEMN-soft, AMFMN-fusion, and AMFMN-sim separately.

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TABLE I
RETRIEVAL RESULTS OF THE TRADITIONAL CROSS-MODAL METHODS ON RSICD AND RSITMD

Method	RSICD dataset							RSITMD dataset						
	Sentence Retrieval			Image Retrieval			mR	Sentence Retrieval			Image Retrieval			mR
	$R@1$	$R@5$	$R@10$	$R@1$	$R@5$	$R@10$		$R@1$	$R@5$	$R@10$	$R@1$	$R@5$	$R@10$	
AMFMN-soft[13]	6.89	19.94	30.25	5.22	19.73	31.94	18.99	12.46	31.11	44.09	10.61	36.41	55.23	31.65
AMFMN-fusion[13]	7.65	19.69	30.76	5.09	19.35	31.62	19.02	13.12	30.82	42.91	10.72	35.95	53.07	31.10
AMFMN-sim[13]	6.98	20.40	30.12	5.87	20.36	33.09	19.47	12.68	30.52	42.84	9.89	36.36	55.97	31.38
LW-MCR(b)[29]	3.75	18.02	29.92	3.32	14.31	27.30	16.10	12.83	27.21	37.39	9.69	34.60	53.98	29.28
LW-MCR(d)[29]	5.21	17.29	27.99	4.81	18.29	31.71	17.55	13.27	28.98	39.60	10.57	34.46	55.22	30.34
GaLR[12]	6.61	19.36	30.00	5.03	20.01	33.32	19.06	13.12	30.45	42.91	10.87	35.07	53.12	30.92
MSIT[34]	6.46	23.69	38.45	4.96	20.74	35.86	21.69	14.37	33.18	48.74	9.34	37.62	58.81	33.67
IEFT[11]	8.11	27.45	43.25	6.26	24.00	39.98	24.84	15.48	37.61	51.40	11.10	37.99	58.75	35.38
GLISA	**20.68**	**40.62**	**53.34**	**14.73**	**40.22**	**56.52**	**37.69**	**32.08**	**51.99**	**63.94**	**23.36**	**58.27**	**74.47**	**50.69**

TABLE II
RETRIEVAL RESULTS OF THE CLIP-BASED METHODS ON RSICD AND RSITMD

Method	RSICD dataset							RSITMD dataset						
	Sentence Retrieval			Image Retrieval			mR	Sentence Retrieval			Image Retrieval			mR
	R@1	R@5	R@10	R@1	R@5	R@10		R@1	R@5	R@10	R@1	R@5	R@10	
CLIP[26]	13.54	30.83	43.46	11.55	33.14	49.83	30.39	24.16	47.12	61.28	20.40	50.53	68.54	45.33
Single Language[32]	10.70	29.64	41.53	9.14	28.96	44.59	27.42	19.69	40.26	54.42	17.61	49.73	66.59	41.38
CoOp[54]	6.32	15.89	27.60	4.31	17.89	31.73	17.29	12.19	30.69	42.82	9.16	33.85	54.35	30.51
VPT[62]	7.23	19.40	30.77	6.94	25.26	40.73	21.72	14.98	32.05	40.15	15.97	41.35	60.35	34.14
Adapter[55]	8.73	24.73	37.81	8.43	26.02	43.33	24.84	13.75	27.64	39.96	12.89	40.09	59.91	32.37
CLIP-Adapter[56]	7.11	19.48	31.01	7.67	24.87	39.73	21.65	12.83	28.84	39.05	13.30	40.20	60.06	32.38
Convpass[57]	6.54	19.67	32.78	7.03	23.08	39.15	21.38	16.03	30.16	40.26	12.05	38.66	58.11	32.55
AdaptFormer[58]	12.46	28.49	41.86	9.09	29.89	46.81	28.10	16.71	30.16	42.91	14.27	41.53	61.46	34.81
Cross-Modal Adapter[59]	11.18	27.31	40.62	9.57	30.74	48.36	27.96	18.16	36.08	48.72	16.31	44.33	64.75	38.06
UniAdapter[60]	12.65	30.81	42.74	9.61	30.06	47.16	28.84	19.86	36.32	51.28	17.54	44.89	65.46	39.23
PETL[35]	14.13	31.51	44.78	11.63	33.92	50.73	31.12	23.67	44.07	60.36	20.10	50.53	67.97	44.47
GLISA	**19.52**	**40.44**	**52.28**	**14.75**	**39.50**	**55.46**	**36.99**	**28.21**	**50.94**	**62.50**	**20.50**	**54.80**	**72.19**	**48.41**

2) *LW-MCR [29]:* LW-MCR is a lightweight multiscale crossmodal retrieval model, which is also used to solve the problems of multiscale scarcity and target redundancy in CMRSITR. In addition, it uses knowledge distillation to further improve retrieval performance. The main difference between LW-MCR(b) and LW-MCR(d) is whether or not AMFMN is used as a teacher network in the knowledge distillation.
3) *GaLR [12]:* GALR uses global and local information of RS images for cross-modal retrieval, which propose an MIDF module to dynamic fuse these information. Besides, an MR algorithm is proposed to optimize the retrieval results.
4) *MSIT [34]:* MSIT considers the inherent relationship between different modal features and proposes a lightweight multiscale feature learning module to extract more useful information.
5) *IEFT [11]:* IEFT regards images and texts as a whole and finds the inherent relationship between them. IEFT not only narrows the gap between modalities but also enhances visual features. Finally, the RM is used to output the matching scores.

Table I presents the experimental results of these methods on different datasets, and we will analyze and discuss them below.

1) *Results on the RSICD Dataset:* We first evaluate our method on the large-scale CMRSITR dataset RSICD. The left of Table I shows the retrieval results of our method and other methods on the RSICD dataset. Our proposed GLISA method achieves 37.69% mR, 20.68% $R@1$ of sentence retrieval, and 14.73% $R@1$ of image retrieval, respectively. Compared with the DCNN-based methods (AMFMN, LW-MCR, and GALR), the retrieval results of our GLISA method have better performance. Compared with the GALR method that uses GaLR to improve retrieval performance, our experiment result mR has improved by 18.63%, $R@1$ of sentence retrieval has improved by 14.07%, and $R@1$ of image retrieval

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has improved by 9.7%. Furthermore, compared with the transformer-based methods (MSIT and IEFT), our method also outperforms them. Our GIISA significantly surpasses the recent state-of-the-art method IEFT by 12.39%, 12.57%, and 8.47% on mR, $R@1$ of sentence retrieval, and $R@1$ of image retrieval, respectively. Moreover, we find that the transformer-based methods has a better performance than the DCNN-based methods.

2) *Results on the RSITMD Dataset:* Compared with the RSICD dataset, the RSITMD dataset has low intraclass similarity. The right of Table I shows the retrieval results of our method and other methods on the RSITMD dataset. Our proposed GLISA method reaches 50.69%, 32.08%, and 23.36% with mR, $R@1$ of sentence retrieval, and $R@1$ of image retrieval, respectively, and achieves the best performance. Compared with the GALR method, the GLISA method improves 19.77%, 18.96%, and 12.49% on mR, $R@1$ of sentence retrieval, and $R@1$ of image retrieval, respectively. Compared with the previous best method IEFT, our method shows an improvement of 15.31%, 16.57%, and 12.26% on mR, $R@1$ of sentence retrieval, and $R@1$ of image retrieval, respectively.

2) CLIP-Based Methods: Following [35], we compare our method with several CLIP-based methods, including full fine-tuning [26], [32], prompt learning [54], [62], and adapter tuning [55], [56], [57], [58], [59], [60]. The results of this comparison are presented in Table II.

1) *Results on the RSICD Dataset:* As shown on the left side of Table II, the retrieval performance of our method on the RSICD dataset outperforms the other CLIP-based methods. Compared with the state-of-the-art method PETL, our method improves 5.87%, 5.39%, and 3.12% on mR, $R@1$ of sentence retrieval, and $R@1$ of image retrieval, respectively.
2) *Results on the RSITMD Dataset:* The results presented on the right side of Table II demonstrate that our method achieves superior retrieval performance on the RSITMD dataset compared with o other CLIP-based methods. When compared with the state-of-the-art method CLIP, our approach shows improvements of 3.08% and 4.05% in terms of mR and $R@1$ for sentence retrieval, respectively.

According to the above results, by focusing on both global and local alignment, we achieve a comprehensive and robust retrieval framework that surpasses the existing methods in terms of performance and adaptability.

C. Ablation Studies

As shown in Fig. 2, our proposed GLISA method consists of a GIE module, an ALIE module, and an LISA model. In this section, we conduct extensive ablation studies on the RSITMD dataset to analyze the impact of these modules.

First, we designed eight sets of experiments to combine these modules and optimize them with different loss functions on the RSITMD dataset.

1) *No.1:* Only use the GIE module to extract global information, and optimize it with a bidirectional triplet loss $\mathcal{L}_{\text{tri},g}$ [13].
2) *No.2:* Use the GIE module to extract global information, and optimize it with $\mathcal{L}_{\text{itc},g}$.
3) *No.3:* Use the GIE and ALIE modules to extract GaLR separately, optimize GIE with $\mathcal{L}_{\text{itc},g}$, and optimize ALIE with $\mathcal{L}_{\text{itc},l}$ and $\mathcal{L}_{\text{div}}$.
4) *No.4:* Use the GIE and ALIE modules to extract GaLR separately, optimize GIE with $\mathcal{L}_{\text{itc},g}$, and optimize ALIE with $\mathcal{L}_{S-\text{itc},l}$ and $\mathcal{L}_{\text{div}}$. $\mathcal{L}_{S-\text{itc},l}$ denotes using label smoothing to soften the one-hot targets.
5) *No.5:* Use the GIE and ALIE modules to extract GaLR separately, optimize GIE with $\mathcal{L}_{\text{itc},g}$, and optimize ALIE with $\mathcal{L}_{A-\text{itc},l}$ and $\mathcal{L}_{\text{div}}$. $\mathcal{L}_{A-\text{itc},l}$ indicates that $p_{i,j,k}(V_l, V_l)$ or $p_{i,j,k}(T_l, T_l)$ is computed using $\text{sim}(\cdot,\cdot)$ instead of $\widetilde{\text{sim}}(\cdot,\cdot)$.
6) *No.6:* Only use the ALIE module to extract local information, and optimize it with $\mathcal{L}_{\text{soft}-\text{itc},l}$ and $\mathcal{L}_{\text{div}}$.
7) *No.7:* Use the GIE and ALIE modules to extract GaLR separately, optimize GIE with $\mathcal{L}_{\text{itc},g}$, and optimize ALIE with $\mathcal{L}_{\text{soft}-\text{itc},l}$.
8) *No.8:* Use the GIE and ALIE modules to extract GaLR separately, optimize GIE with $\mathcal{L}_{\text{itc},g}$, and optimize ALIE with $\mathcal{L}_{\text{soft}-\text{itc},l}$ and $\mathcal{L}_{\text{div}}$.

Table III shows the results of the ablation studies, and we will compare these results and analyze them below.

1) *No.1 Versus No.2:* Compared with No.1, No.2 shows an improvement of 2.38% on mR. This demonstrates the effectiveness of the ITC loss $\mathcal{L}_{\text{itc},g}$ and highlights that $\mathcal{L}_{\text{itc},g}$ does not require margin setting like the bidirectional triplet loss $\mathcal{L}_{\text{tri},g}$.
2) *No.2 Versus No.3:* When the ALIE module was added to extract local information, $R@1$ of sentence retrieval is improved by 2.65%, $R@10$ of image retrieval is improved by 3.45%, and mR is improved by 0.69%. The comparison of No.2 and No.3 illustrates that the local information alignment of image–text pairs is crucial for CMSITR. However, when using $\mathcal{L}_{\text{itc},l}$ for local information alignment, there can be semantic ambiguity, leading to a lesser performance improvement.
3) *No.3 Versus No.4:* When $\mathcal{L}_{S-\text{itc},l}$ is used to optimize the ALIE module instead of $\mathcal{L}_{\text{itc},l}$, $R@1$ of sentence retrieval, $R@1$ of image retrieval, and mR are improved by 2.88%, 0.35%, and 0.85%, respectively. This indicates that using label smoothing can alleviate semantic ambiguity to some extent.
4) *No.3 Versus No.5:* Compared with No.3, the mR of No.5 is improved by 0.85%, which is almost the same as the mR of No.4. This indicates that $\mathcal{L}_{A-\text{itc},l}$ can also alleviate semantic ambiguity.
5) *No.3 Versus No.8:* Compared with No.3, No.8 has a 4.73% improvement on $R@1$ of sentence retrieval, a 1.41% improvement on $R@1$ of image retrieval, and a 1.6% improvement on mR. Compared with No.4 and No.5, No.8 has better retrieval performance, indicating the effectiveness of the LISA module.

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TABLE III
ABLATION STUDIES ON DIFFERENT COMPONENTS OF GLISA ON RSITMD

No.	GIE		ALIE		LISA			Sentence Retrieval			Image Retrieval			mR
	$\mathcal{L}_{tri,g}$	$\mathcal{L}_{itc,g}$	$\mathcal{L}_{itc,l}$	$\mathcal{L}_{div}$	$\mathcal{L}_{S\text{-}itc,l}$	$\mathcal{L}_{A\text{-}itc,l}$	$\mathcal{L}_{soft\text{-}itc,l}$	$R@1$	$R@5$	$R@10$	$R@1$	$R@5$	$R@10$	
1	✓							24.56	48.45	60.40	20.75	52.39	69.56	46.02
2		✓						25.00	51.99	**65.71**	21.64	55.22	70.84	48.40
3		✓	✓	✓				27.65	49.78	63.72	21.95	57.12	74.29	49.09
4		✓		✓	✓			30.53	51.99	63.94	22.30	57.65	73.23	49.94
5		✓		✓		✓		28.76	51.99	64.60	22.21	57.83	74.07	49.91
6				✓			✓	25.66	46.46	58.19	20.62	54.82	72.04	46.30
7		✓					✓	30.53	49.78	62.83	22.35	57.74	74.07	49.55
8		✓		✓			✓	**32.08**	**51.99**	63.94	**23.36**	**58.27**	**74.47**	**50.69**

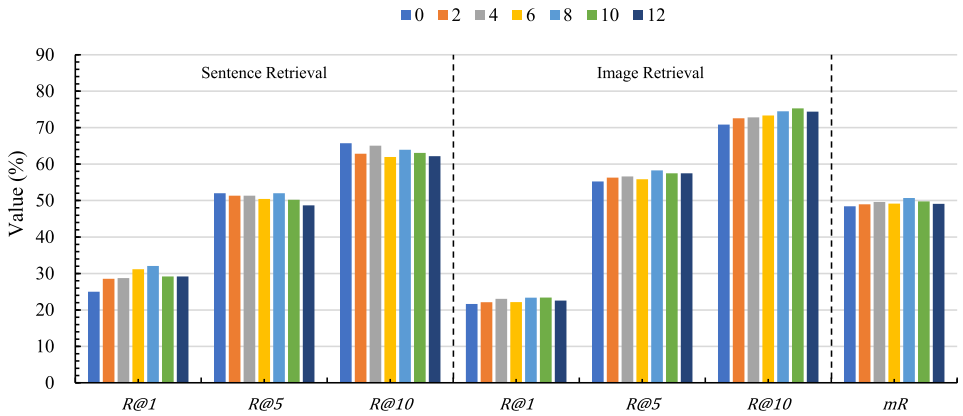

Fig. 6. Illustration of the influence of the number of heads K on $R@k$ and mR of the RSITMD dataset. The value of K is selected with an interval of 2 from 0 to 12.

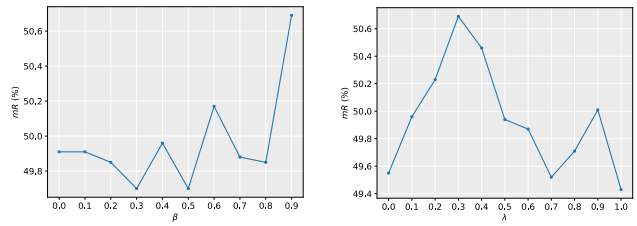

Fig. 7. (Left) Ablations of the parameter β on the RSITMD dataset. (Right) Ablations of the loss balance factor λ on the RSITMD dataset.

6) *No.2 Versus No.8:* By adding the ALIE and LISA modules to the GIE module, $R@1$ of sentence retrieval, $R@1$ of image retrieval, and mR are improved by 7.08%, 1.72%, and 2.29%, respectively. Due to the presence of numerous targets and complex background information in RS images, extracting local information of the image–text pairs and aligning them can effectively enhance retrieval performance.
7) *No.6 Versus No.8:* When the GIE module is removed from the GLISA model, $R@1$ of sentence retrieval is decreased by 6.42%, $R@1$ of image retrieval is decreased by 2.74%, and the mR is decreased by 4.39%. Global information is essential for capturing relationships between targets, and thus removing the GIE module results in a degradation of retrieval performance.
8) *No.7 Versus No.8:* To further verify the effectiveness of diversity loss $\mathcal{L}_{\text{div}}$, the GLISA model was optimized without $\mathcal{L}_{\text{div}}$ (No.7). Compared with No.8, $R@1$ of sentence retrieval, $R@1$ of image retrieval, and mR are decreased by 1.55%, 1.01%, and 1.14% separately. When $\mathcal{L}_{\text{div}}$ is removed from the GLISA model, the local information extracted by the ALIE module lacks diversity, resulting in information redundancy.

Then, we analyze the influence of the number of heads K on the retrieval performance of the GLISA model. Fig. 6 illustrates the different results for different values of K. It can be observed that as K increases, $R@1$ of sentence retrieval, $R@1$ of image retrieval, and mR initially increase and then decrease. When K is small, the local information of image–text pairs is not fully explored. However, when K is large, irrelevant noise is introduced, which can negatively impact retrieval performance.

Finally, we analyze the effects of the parameter β and the loss balance factor λ on the experimental results in Fig. 7. First, for the parameter β, the best performance is achieved when β is 0.9. The retrieval performance is poor when β is small. It is possible that there is some noise during the initial training phase of the model. Second, as the loss balance factor λ increases, mR initially increases and then decreases. The proposed method with $\lambda = 0.3$ performs the best result.

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1. The gray bridge is empty with houses and cars on both sides.
2. The gray bridge is empty with houses and cars on both sides.
3. There is a twin-tower bridge on the other side of the river.
4. There is a bridge on the river and there are buildings near it.
5. The bridge is on the river and the boat is on the river.
6. The gray bridge is empty, with houses and cars on both sides.
7. There is a twin-tower bridge across the river.
8. There is a bridge on the river and there are some buildings near it.
9. A gray bridge was empty, with houses and cars on both sides.
10. The gray bridge was empty, with houses and cars on both sides.

1. Beside a bridge covered with white iron, there was a moving boat.
2. Beside a bridge full of white iron, a boat was moving.
3. A small boat sailed by a bridge full of white iron.
4. Beside a bridge full of white iron, there was a moving boat.
5. There is a twin-tower bridge across the river.
6. There is a twin-tower bridge on the other side of the river.
7. Beside a bridge, full of white iron, there was a movable boat.
8. The bridge is on the river and the boat is on the river.
9. The bridge is on the river and the boat is in the river.
10. The bridge is on the river and the boat is in the river.

(a)

1. The baseball field is in the corner of the coressed haighway.
2. The gray house met with green plants beside the red and yellow baseball field.
3. A baseball field was built on the lawn, and the same building stood straight.
4. The fanshaped baseball field in a square ground.
5. The gray house met the green plants beside the red and yellow baseball field.
6. The baseball field is built on the lawn, and the same building is on the straight side.
7. The gray house found the green plants beside the red and yellow baseball field.
8. There is a fan shaped baseball field near a building looking like a boomerang.
9. A building with a tree is near a baseball field.
10. Four baseball field in every quarter of a circle lawn.

1. The gray house met the green plants beside the red and yellow baseball field.
2. The gray house met with green plants beside the red and yellow baseball field.
3. The gray house found the green plants beside the red and yellow baseball field.
4. A building with a tree is near a baseball field.
5. There is a fan shaped baseball field near a building looking like a boomerang.
6. This fan shaped baseballfield is surrounded by roads and a building.
7. The baseball field is built on the lawn, and the same building is on the straight side.
8. A baseball field was built on the lawn, and the same building stood straight.
9. The baseball field is in the corner of the coressed haighway.
10. There is a gray room next to a baseball field.

(b)

Fig. 8. Comparison of top-ten retrieved results on the RSITMD dataset between GIE and GLISE for each image query. The matched and mismatched texts are marked with green and black, respectively. (a) Results of sentence retrieval about bridge. (b) Results of sentence retrieval about baseball field.

D. Qualitative Results

Fig. 8 illustrates the top-ten sentence retrieval results of GIE and GLISA for the given image query where we use the highlighted words and image region boxes to indicate their correspondence. As shown in Fig. 8(a), even when GIE fails to retrieve the correct sentence results, GLISA can still retrieve all the correct results in the top-ten. We find that GLISA can focus on more local semantic regions, such as bridge and boat. Besides, we also noted that GLISA seems to understand the material of the bridge and the movement of the boat. As presented in Fig. 8(b), although both GIE and GLISA can retrieve three correct sentence results in the top-ten, the correct results of GLISA are ranked higher, appearing in the top-three.

Fig. 9 shows the top-ten image retrieval results of GIE and GLISA for a given text query. As shown in Fig. 9(a), although the first three image retrieval results of GIE include the targets mentioned in the text query, such as train station, house, and building, they do not match the building texture and the number of buildings mentioned in the caption. In contrast to GIE, GLISA can retrieve the correct image that matches the text query in the top-one. As depicted in Fig. 9(b), although both GIE and GLISA can identify the presence of the gray port and the ships in the RS images, GLISA can discover the relationship between the large ships and the red ships, as well as the quantity of the large ships.

Based on the above, GIE only focuses on global information, which leads to the neglect of crucial local information in the process of image–text matching. The proposed GLISA method not only fully exploits the fine-grained discriminative cues to determine the matching between RS images and texts but also discovers the texture of the targets, the number of targets, and the relationship between the targets. Therefore, local information alignment is important for the CMRSITR task to effectively improve retrieval performance.

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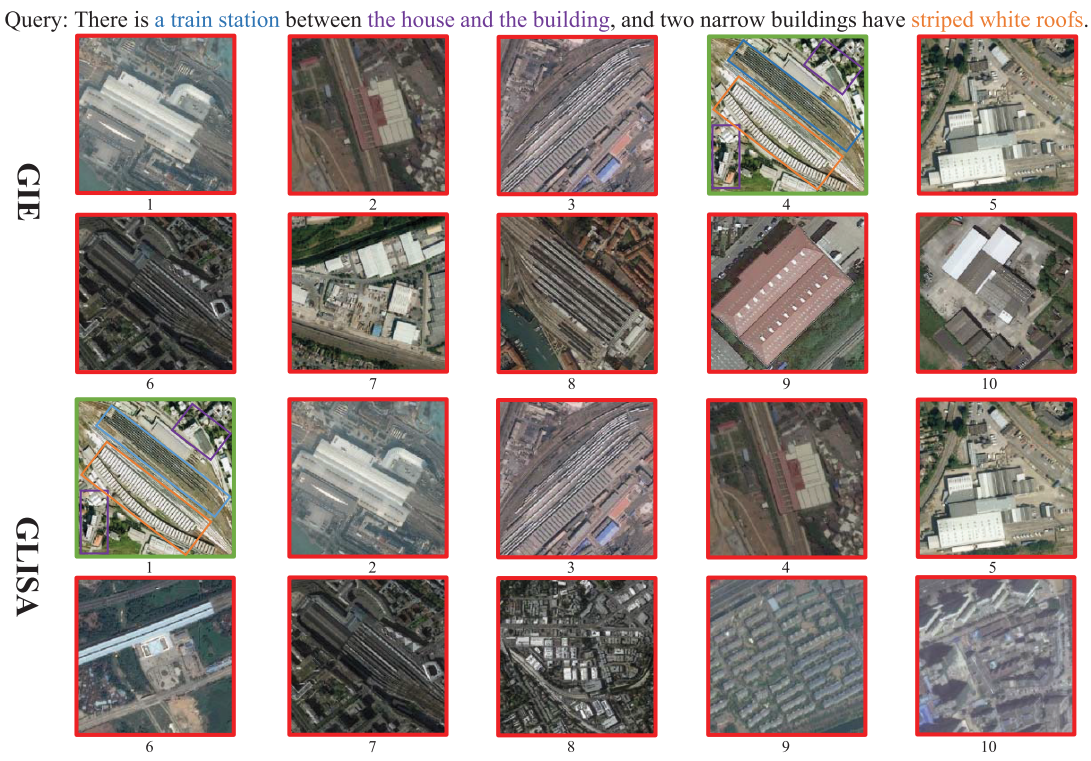
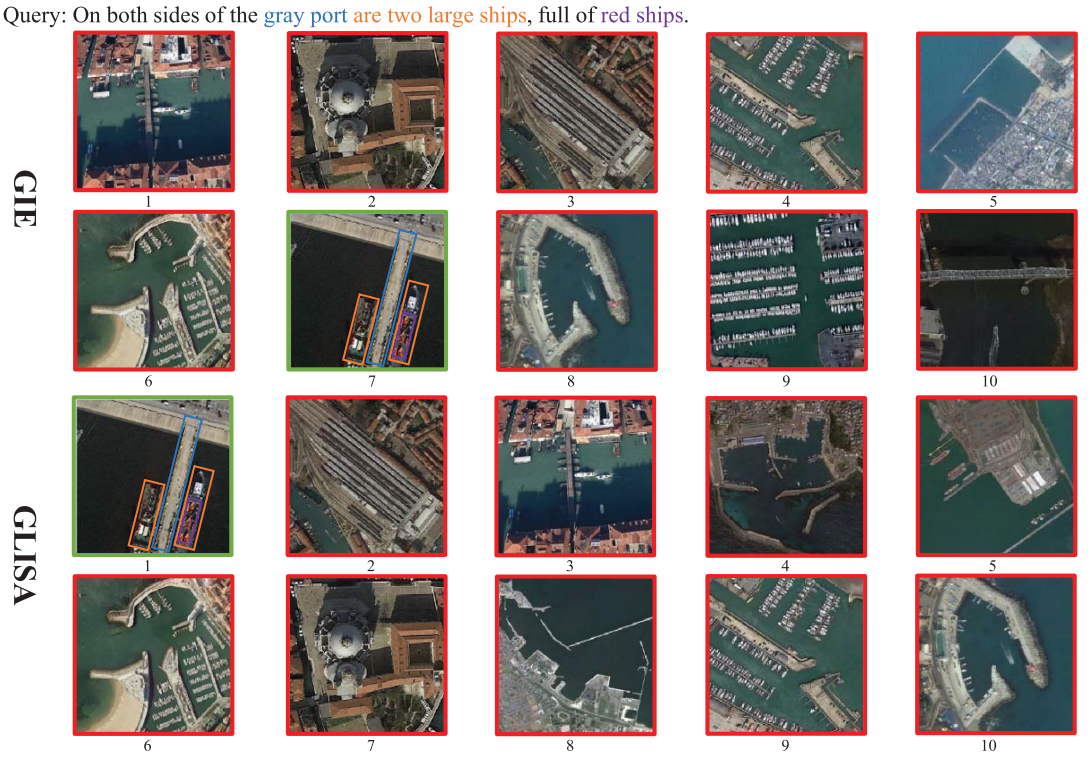


(a)



(b)

Fig. 9. Comparison of top-ten retrieved results on the RSITMD dataset between GIE and GLISE for each text query. The matched and mismatched RS images are marked with green and red rectangles, respectively. (a) Results of image retrieval about railway station. (b) Results of image retrieval about port.

V. Conclusion

This article introduces the GLISA framework for CMR-SITR. It effectively learns discriminative global image–text features while incorporating local region-word information. Through the utilization of the GIE module, the model successfully extracts global features and aligns them using the ITC loss. The ALIE module adaptively mines local

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representations, capturing fine-grained details and context-specific information. Furthermore, the LISA module aligns these local representations, mitigating semantic ambiguities. Experimental evaluations on the CMRSITR datasets demonstrate the superior retrieval performance of the GLISA model, highlighting its effectiveness in the cross-modal retrieval task. The study also establishes the importance of aligning local information for CMRSITR. Overall, the GLISA model presents a promising approach for enhancing cross-modal retrieval in RS image–text datasets. Future research directions may involve exploring additional techniques to further improve the alignment and retrieval performance for more cross-modality including SAR and infrared RS image.

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